

Heat Transfer in Plate Heat Exchangers

Plate heat exchangers are constructed from individual pre-formed, patterned plates clamped together. The gaps between the plates are called channels and the process fluids flow in alternate channels in a counter-flow arrangement. These plates may be arranged to give one or two pass combinations and also more that one exchanger can be located on the same frame.

The gap between the plates is small (2 - 5mm) and the corrugations further reduce the flow area. The high turbulence caused by the corrugations lead to high heat transfer coefficients and this coupled by the high surface areas of the plates gives high heat transfer.

Heat Transfer Coefficient from Kumar (1964)

$$Nu = C Re^y Pr^{0.33} \left(\frac{\mu_b}{\mu_s} \right)^{0.17}$$

$$\text{where } Re = \frac{\dot{m} d_e}{A_{xsa} \mu_b}$$

$\dot{m}$ mass flow rate per channel (kg/m²s)
 A_{xsa} flow area (m²)
 d_e hydraulic diameter (m)
 μ_b viscosity at bulk temperature (Ns/m)

The values of C and y are found from the table opposite

It should be noted that turbulence occurs at very small Reynolds Numbers

Plate Geometry (R_L)

Path length: to account for the increased flow path created by the corrugations a correction factor is used:

$$R_L = \text{developed length} / \text{projected length}$$

This value is normally between 1.1 and 1.25 and an average value of 1.17 can be used.

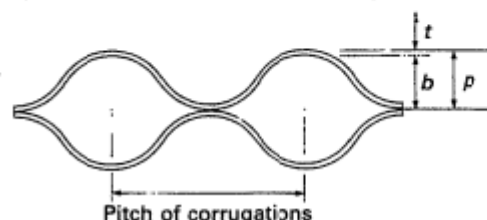
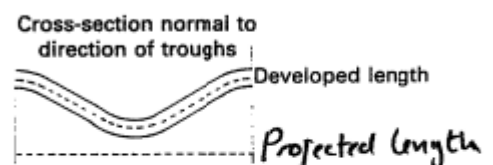
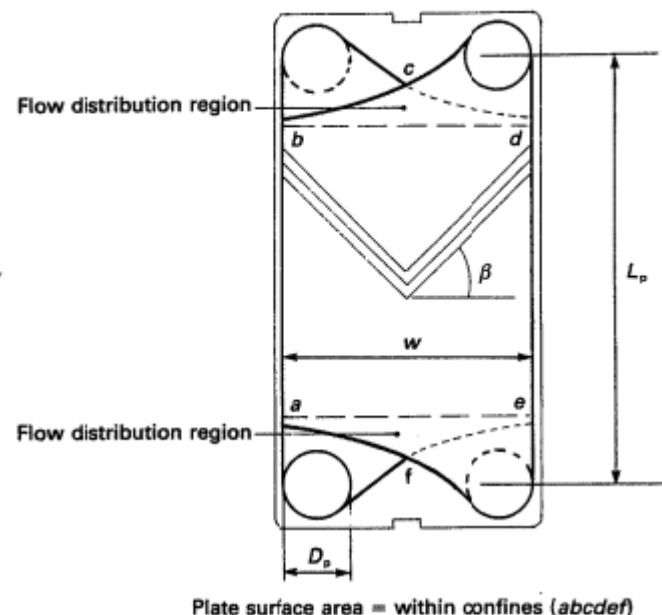
Mean Channel Gap (b)

The mean channel (flow) gap is given by $b = p - t$

where p = pitch (m)
 t = thickness of plate (m)

(Note that the pitch here is an expression of the depth of the corrugations.)

Chevron angle (degrees)	Heat transfer		
	Reynolds number	C	y
≤ 30	≤ 10	0.718	0.349
	> 10	0.348	0.663
45	< 10	0.718	0.349
	10 – 100	0.400	0.598
	> 100	0.300	0.663
50	< 20	0.630	0.333
	20 – 300	0.291	0.591
	> 300	0.130	0.732
60	< 20	0.562	0.326
	20 – 400	0.306	0.529
	> 400	0.108	0.703
≥ 65	< 20	0.562	0.326
	20 – 500	0.331	0.503
	> 500	0.087	0.718



Channel Flow Area (A_{xsa}) $A_{xsa} = b w$

where w = effective plate width (m)

Hydraulic Diameter (d_e)

$$d_e = \frac{4 (\text{channel flow area})}{(\text{wetted surface})} = \frac{4 A_{xsa}}{P}$$

and as $P = 2 (b + R_L w)$

$$d_e = \frac{4 b w}{2 (b + R_L w)}$$

typically b is $\ll$ less than w , hence $d_e = 2 b / R_L$ (or $d_e = 2 b / 1.17$)

Heat Transfer from Plates

The LMTD and Effectiveness approaches can be used.

LMTD Correction is required as the end plates have heat transfer from one side only and in a multi-pass arrangement the middle plates may be in parallel flow. However if the number of plates exceed 20 these effects can be ignored.

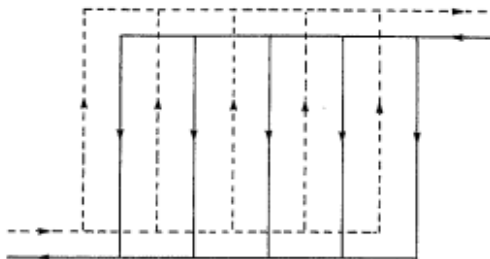
Overall Heat Transfer Coefficient:

$$\frac{1}{U} = \frac{1}{h_1} + r_{f1} + \frac{x}{k} + r_{f2} + \frac{1}{h_2}$$

where r_{f1} and r_{f2} are the appropriate fouling factors.

Flow rate per Channel

The diagram shows that in a single pass, counter-flow heat exchanger there are 11 plates forming 10 channels. The process fluids flow through 5 channels and the end plates have heat transfer on one side only.



Single pass
(counter-flow)



two pass - two pass
(counter-flow except middle plates)